Secure Pseudonymetry with DSSS Watermarking for LEO Satellites

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Problem:

Dense LEO constellations, interfering signals, and the lack of a direct transmitter identity make reliable signal association difficult.

Our Approach:

A low-power DSSS watermark carries a pseudonymous satellite ID and keyed authentication without requiring payload demodulation.

Why It Matters:

The design supports multi-satellite operation, remains compatible with legacy receivers, and extends to other signals requiring embedded identity and authentication.

Background

Which satellite transmitted the pilot tones?

No direct identity: A received LEO beacon does not directly reveal a readily usable transmitter identity.

Dense constellations: More simultaneously visible satellites increase signal overlap and association ambiguity.

Interference/spoofing: Intra-/inter-constellation interference and spoofing further complicate reliable association.

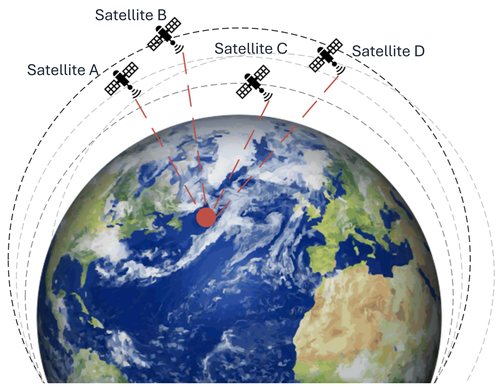


Fig. 1: Dense multi-satellite setting.

What if satellites carry a low-power identity watermark?

Each satellite embeds a weak DSSS watermark carrying its pseudonymous ID and authentication information alongside the existing signal.

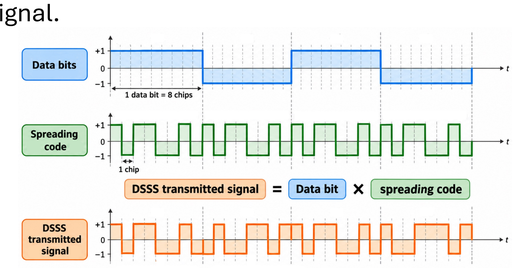


Fig. 2: Direct sequence spread spectrum (DSSS) structure

Low power: minimally intrusive to the host signal

Spread spectrum: robust acquisition with spreading gain and code separation

Backward compatible: legacy receivers can ignore the watermark

Broader Applicability

The DSSS watermark is **not limited to LEO beacons**; it can augment other signals that require embedded identity and authentication.

The low-power overlay enables backward **compatibility** with legacy receivers while supporting enhanced watermark-aware receivers.

Design and Implementation

The design targets future transmitter-assisted deployment while preserving compatibility with legacy receivers.

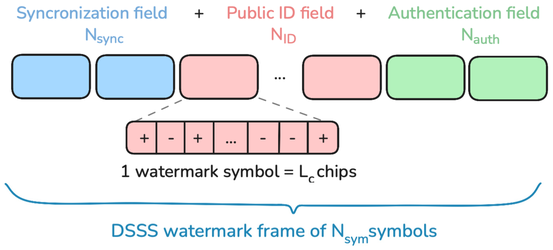


Fig. 3: DSSS watermark frame structure

The DSSS frame carries synchronization, pseudonymous ID, and keyed authentication fields.

Receiver processing:

Correlation-based acquisition → Pseudonymous ID recovery → Keyed authentication

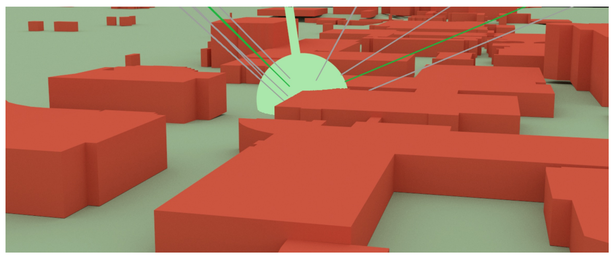


Fig. 4: Sionna Evaluation Environment

Performance is evaluated using link-level simulations in **Sionna** [1]. Ray-traced multi-satellite scenarios are used for acquisition, coexistence, and authentication robustness.

Table 1: Main Parameters

Parameters	
Carrier: 11.325 GHz	Bandwidth: 1MHz
Chip rate: 400 kchip/s	Frame duration: ~0.16 s
Code length: 511	Beacon offset: −25 dB
Samples/chip: 6	Watermark offset: −35 dB

Minimal impact on the host signal

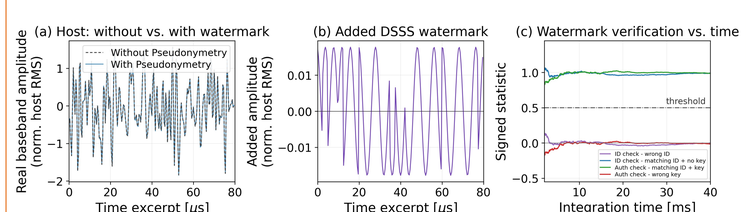


Fig. 5: Time-domain DSSS watermark insertion and verification

Results

Can a weak watermark be acquired?

Reliable acquisition and keyed verification emerge as watermark power increases.

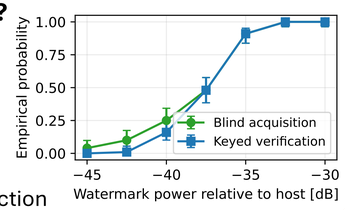


Fig. 6: Watermark detection

Can multiple satellites coexist?

High acquisition and verification remain robust as satellite count increases, and wrong-ID probability drops as the candidate watermark strengthens.

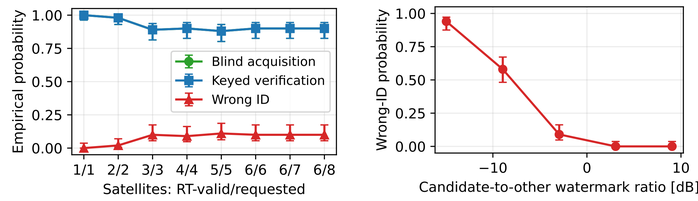


Fig. 7: Multi-satellite coexistence performance

Can authentication distinguish legitimate and spoofed signals?

Incorrect identities are increasingly rejected as the authentic watermark becomes dominant.

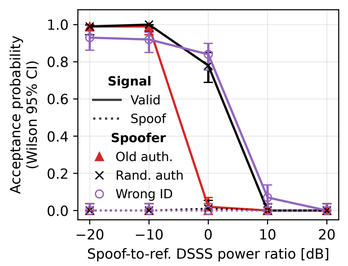


Fig. 8: Authentication stress test

Does the watermark preserve the host signal?

Low-power watermarking introduces limited host-signal distortion at the intended operating point.

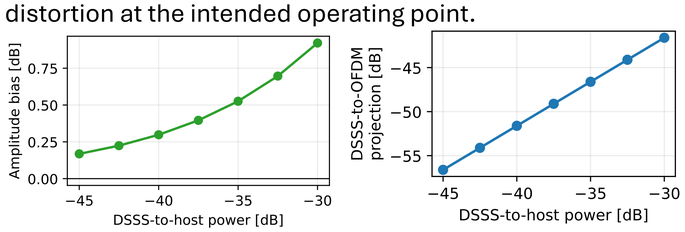


Fig. 9: DSSS watermark effect on existing downlink observables

References

[1] J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller, "Sionna," 2022, https://nvlabs.github.io/sionna/.

[2] N. Camuzcu, T. M. Hoang, and A. Vahid, "Starlink Beacons for Passive LEO-Aided 9D Navigation," 2026, IEEE VTC 2026-Fall.

[3] M. G. Weldegebriel, J. Wang, G. Hellbourg, N. Zhang, and N. Patwari, "Watermarking of OFDM for Pseudonymetry: Analysis and Experimental Results," in 2024 IEEE International Conference on Communications (ICC) Workshops, 2024, pp. 317–322.

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